How-To Library

Informatica

Creating a REST Web Service in the Informatica Developer Tool

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Abstract

You can create an Informatica REST web service that returns data to a web service client in JSON or XML format. The article explains how to define a REST web service in the Developer tool. The REST web service runs a mapping that returns hierarchical data in JSON format to a web service client browser.

Supported Versions

- Informatica Data Services 10.1.1
- Informatica Data Quality 10.1.1

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Overview

An Informatica REST web service is a web service that receives an HTTP request to perform a GET operation. A GET operation retrieves data. The REST request is a simple URI string from an internet browser. The client limits the web service output data by adding filter parameters to the URI.

For example, a web service client sends a request to retrieve order information for a customer. The URI contains a host port, a web service name, and the resource to access in the web service. The URI also contains a customer ID in the request. The customer ID is the resource key. The REST web service retrieves the order information from an orders table. The REST web service returns the data to the client in a JSON file.

The REST response message format is not defined by a WSDL or schema. Define the output format when you define the Informatica REST resource.

REST Web Service Components

An Informatica REST web service has the following components:

Resource

A resource includes the mapping that the REST web service runs and the definition of the response message that the web service returns. The resource also includes a resource ID, which is a key field in the output data. When you define a resource, you define the structure of the output data that the web service returns to the client. A web service can have multiple resources.

Resource mapping

The mapping that returns the data to return to the web service client. A resource mapping does not read the request query. The REST resource mapping contains a Read transformation. The transformation reads a data object in the Model repository to retrieve data to return to the client. By default, you do not have to add a Filter transformation or a Lookup transformation to retrieve the data based on the client query. The REST web service filters the output data after the mapping returns data.

Request message

A request from a web service client to the web service to perform a task. An Informatica web service can perform an HTTP GET method. The request message is a string that contains the name of the web service, the name and network location of the resource to perform the task, and the parameters to filter the output.

Resource ID

A key field that you can search for in the output data. Each key field has a URL in the output data.

Response message

A JSON or XML file that contains the data to return to the web service client. The response message can contain a hierarchy of elements and multiple-occurring data.

Example REST Web Service

Hypostores has a REST web service that returns all the orders for a customer. A salesman can send a request for the orders from a web browser to the web service. The request contains one or more customer numbers. The web service returns each customer name and a list of all the orders for the customer in a JSON file.

The REST web service contains a mapping that reads a Customers table to retrieve the customer name. The mapping contains a Lookup transformation. The Lookup transformation retrieves all the orders for each customer from an Orders table. The mapping contains a REST Output transformation that returns a hierarchical JSON file. The JSON file structure has customer number and customer name in a parent group. The JSON file contains a child group for the orders within customer. The group contains multiple occurring orders. Each order has an order number, price, and order date.

Creating the REST Web Service

Perform the following steps to create the REST web service:

1. Create the REST web service resource.
2. Define the resource mapping to retrieve the data.
3. Configure the output mapping to map data to elements in the output hierarchy.
4. Test the mapping.
5. Deploy the application to a Data Integration Service.
6. Access the web service from a browser.

Step 1. Create the REST Web Service Resource

A REST web service resource contains the definition of the REST web service response message and a data access method to return the response. When you create an Informatica REST web service, you can define the resource from a data object or you can manually define the resource.

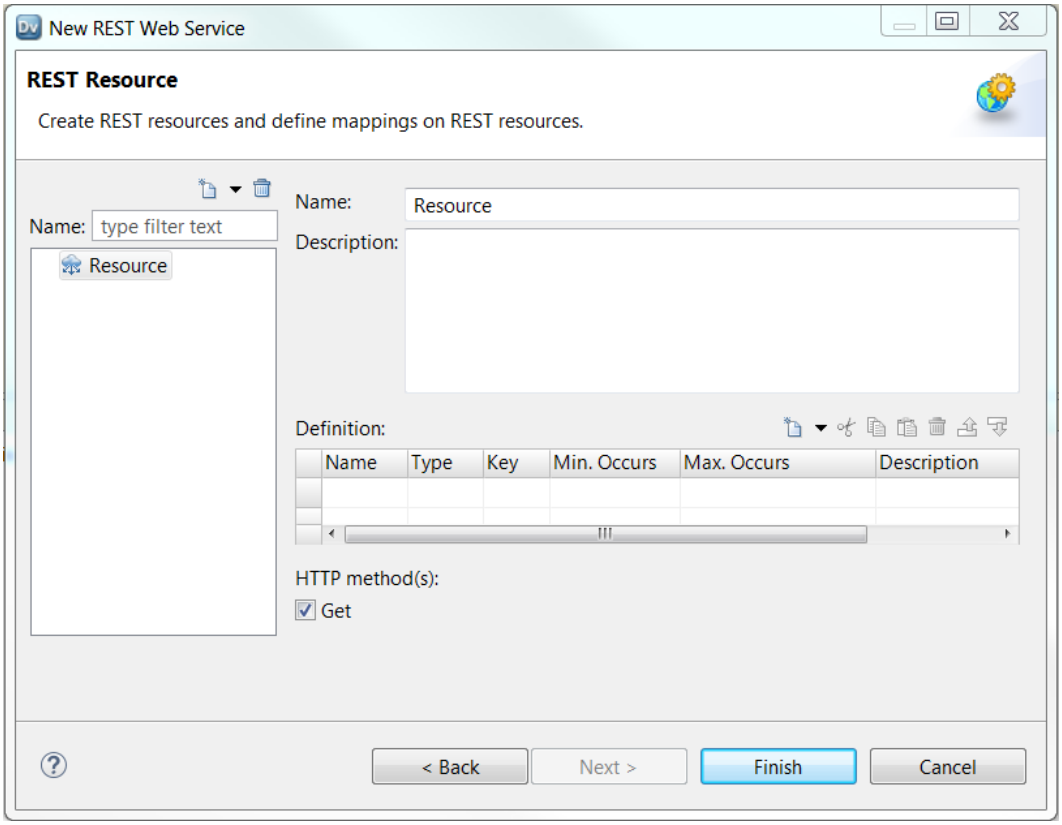
For this example, you manually create the resource because the response message format is hierarchical and it contains multiple-occurring data. You cannot create the resource from a data object in the repository.

Creating the REST Web Service Resource

When you create the web service resource, you define the structure of the response message.

1. In the Developer tool, click **File** > **New** > **Data Service**.
2. Select **REST Web Service** and click **Next**.
3. Enter Orders_Web_Service as the web service name. Click **Next**.
4. In the **REST Resource** dialog box, click **Create From Empty**.

   A default resource appears.



5. In the **Name** field, change the name of the resource from Resource to Orders_Resource.
6. On the **Definition** panel, click **New** > **Element**.
7. Enter the following elements:

Name	Type	Key	Min Occurs	Max Occurs
Customer_Key	string	Yes	1	1

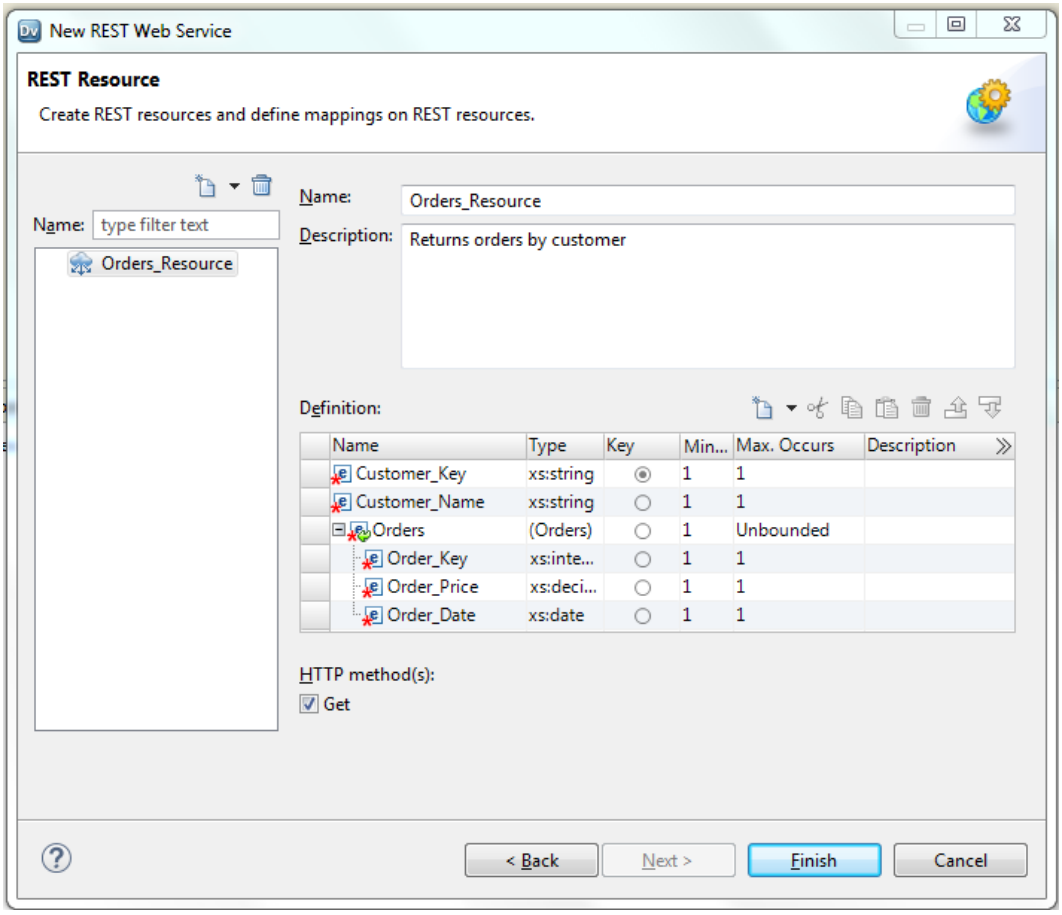
Name	Type	Key	Min Occurs	Max Occurs
Customer_Name	string		1	1
Orders	string		1	unbounded

8. Select the Orders element and click **New** > **Child**.
9. Enter the following child elements beneath Orders:

Name	Type	Key	Min Occurs	Max Occurs
Order_Key	integer		1	1
Order_Price	decimal		1	1
Order_Date	integer		1	1

10. Verify that the Get HTTP method is enabled.
11. Click **Finish**.

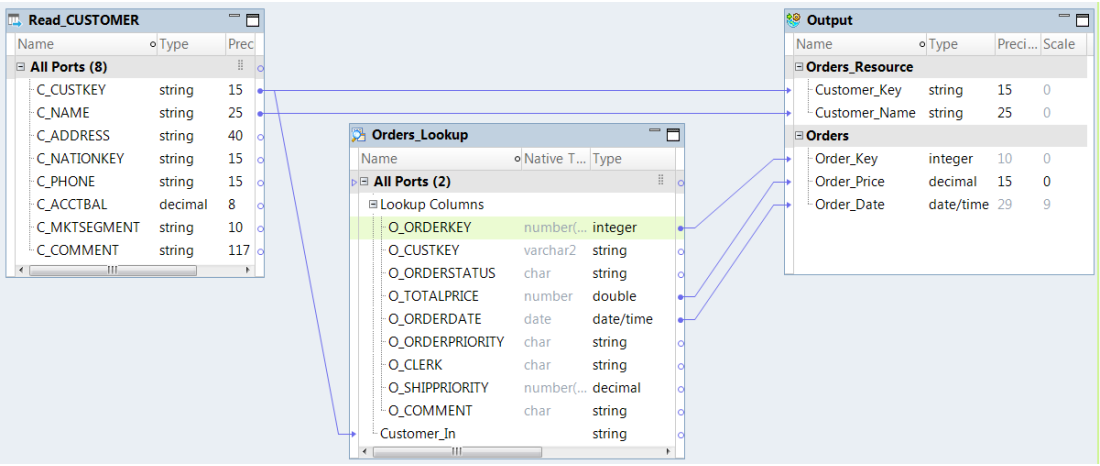
    The following image shows the elements in Orders_Resource:



Step 2. Define the Resource Mapping

After you define the resource, the Developer tool creates a resource mapping that contains a REST Output transformation. The REST Output transformation structure defines the response message that the web service returns to the client. Add a Read transformation and a Lookup transformation to the mapping.

The following image shows the resource mapping to create:



The mapping contains the following objects:

Read_Customer

The Read_Customer transformation reads the customer table. The transformation returns all the customer rows in the table.

Orders_Lookup

The Lookup transformation retrieves the orders for each customer.

Output

The REST Output transformation receives customer information from the Read_Customer transformation. It receives order information from the Orders_Lookup transformation. The REST Output transformation generates a hierarchical JSON file that lists the orders for each customer.

Note: The resource mapping does not contain a Filter transformation to limit the customers to retrieve from the customer table.

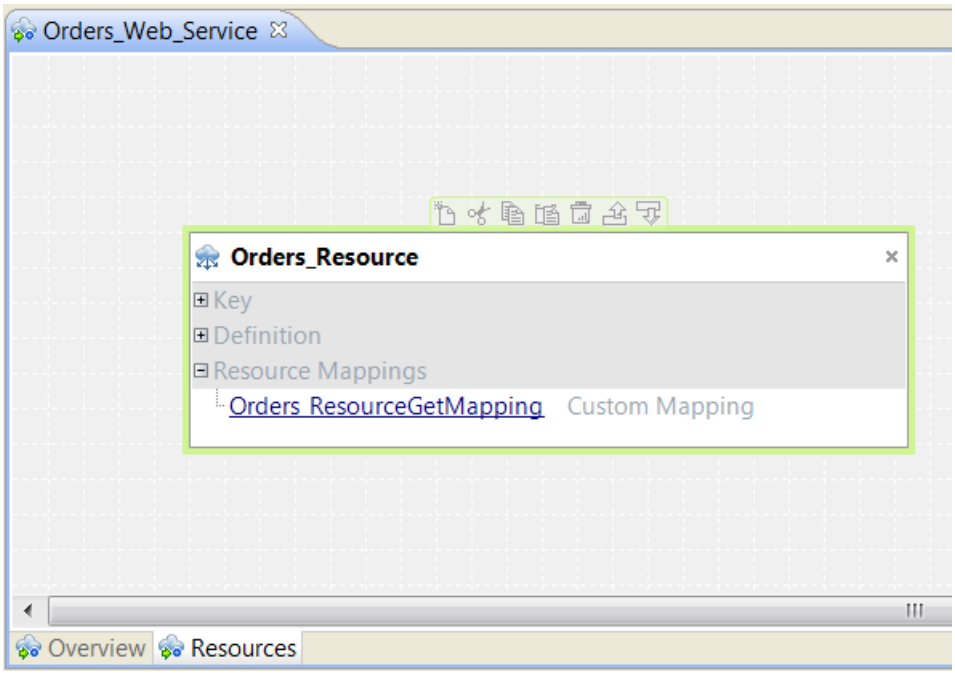
The Data Integration Service filters the output data based on any filter in the client request.

Defining the Resource Mapping

The Developer tool creates a resource mapping that contains the REST Output transformation that you created. Define a Read transformation and a Lookup transformation in the resource mapping.

1. In the **Resources** tab, expand the **Resource Mappings** link in the Orders_Resource.

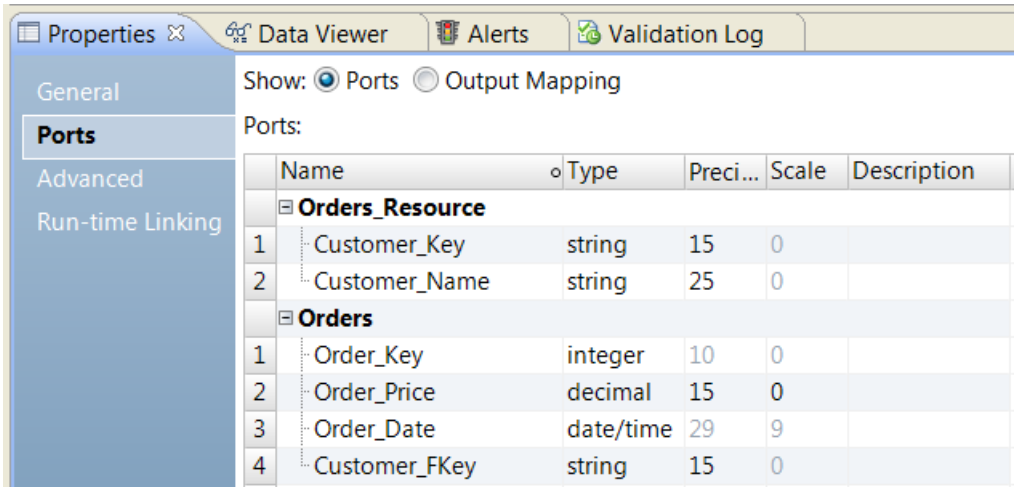
   The following image shows the Orders_Resource link in the Orders_Resource:



2. Click the **Orders_ResourceGetMapping** link in the Orders_Resource.

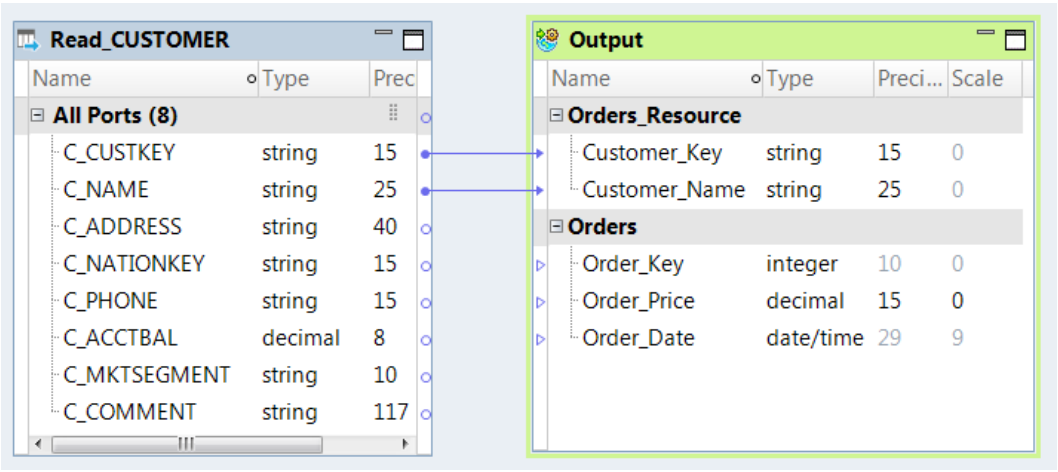
   The mapping appears in the editor.

3. Add the Customer data object to the mapping as a Read transformation.
4. Click the REST Output transformation.
5. On the REST Output transformation **Properties** tab, click the **Ports** view.
6. Change the precision of the Customer_Key to 15 and the precision of the Customer_Name to 25.

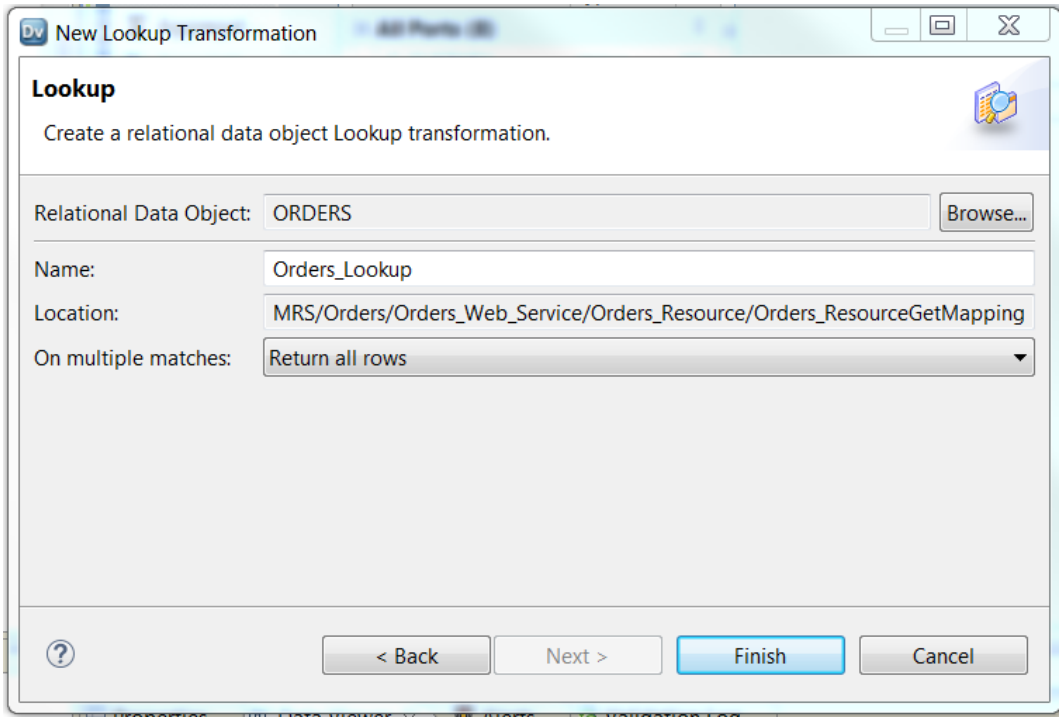


7. Connect the Customer Key and the Customer Name ports from Read_Customer to the REST Output transformation.

   The following image shows the links between the Read_Customer transformation and the REST Output transformation:



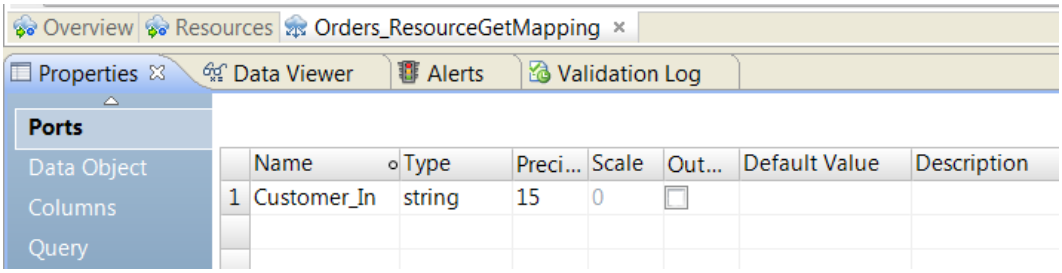
8. To add the Lookup transformation, right-click in the mapping and click **Add Transformation**.
9. Select the Lookup transformation.
10. Choose Relational Data Object Lookup and click **Next**.
11. In the **New Lookup Transformation** dialog box, browse for and select the Orders physical data object. Choose to return all rows.



12. Click **Finish**.

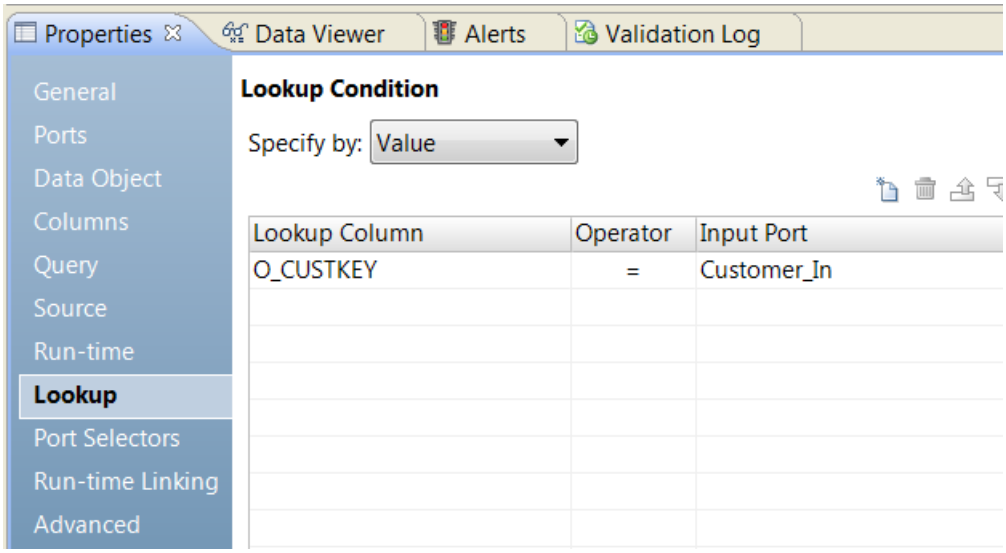
    The Lookup transformation appears in the resource mapping.
13. Click the Lookup transformation in the mapping to select it.

14. In the Lookup transformation **Properties** view, select the **Ports** tab.
15. Add a port called Customer_In. The port is a string with a precision of 15. You do not need to enable it for output.



16. In the Lookup transformation **Properties** view, click the **Lookup** tab.
17. Enter the following lookup condition:

```
O_CUSTKEY = Customer_In
```



18. In the mapping canvas, link the C_CUSTKEY port from Read_Customer to the Customer_In port of the Lookup transformation.
19. Link the following output ports from the Lookup transformation to ports in the REST Output transformation:

Lookup Transformation	Output Transformation
O_ORDERKEY	Order_Key
O_TOTALPRICE	Order_Price
O_ORDERDATE	Order_Date

Step 3. Configure the Output Mapping

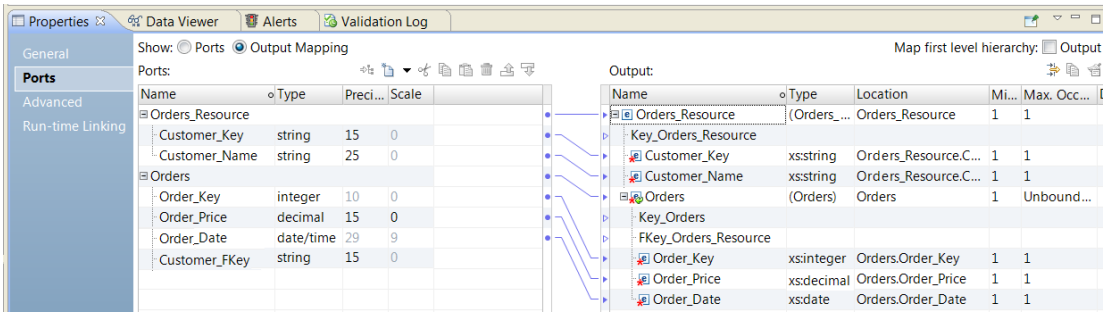
The REST Output transformation receives data from the Customer and Orders transformations and it returns a JSON file.

In the Orders_Resource mapping, the REST Output transformation contains two groups of output data. The parent group contains customer information and it occurs once for each customer. The child group contains the orders. The orders group is multiple-occurring.

When the REST Output transformation contains more than one group of ports, the Developer tool creates keys to link the groups. The Developer tool creates a primary key in each group. The Developer tool creates a foreign key for each child group. The foreign key in the Orders group contains the customer number for each order.

When you view the ports in the REST Output transformation, you do not see the keys. The keys do not have ports. However, you must link data to the keys to define the key values. To view the keys, show the output mapping in the **Ports** view of the REST Output transformation **Properties** tab. The **Ports** view shows the input ports or the output mapping depending on which option you choose to show.

The following image shows the **Output Mapping** view:



The Developer tool creates the following keys in the output:

Key Name	Description
Key_Orders_Resource	The primary key for the Orders_Resource group.
Key_Orders	The primary key for the Orders group.
FKey_Orders_Resource	The foreign key to link the Orders group to the Orders_Resource group.

You must link input ports to the output keys or the mapping is not valid.

Note: You can link a port to more than one element on the **Output** panel. However, you cannot link a field from one group on the **Ports** panel to an element in more than one group on the **Output** panel. All fields in the same **Ports** group must link to elements in the same **Output** group.

Configuring the Output Mapping

To create a primary-key foreign-key relationship between customers and orders, each group must contain the customer number.

1. Add a Customer_Fkey port below Order_Date in the Orders group of the REST Output transformation.

   The Customer_Fkey port receives a customer number. You need to add this port to receive a customer number in the Orders group and to populate the foreign key. The customer number does not appear with each order in the output.

2. Link the Customer_Key from the **Ports** panel to the Key_Orders_Resource on the **Output** panel. To create a link, click the Customer_Key to select it, then select the Key_Orders_Resource. Click the **Map** icon on the **Ports** panel to create the link.

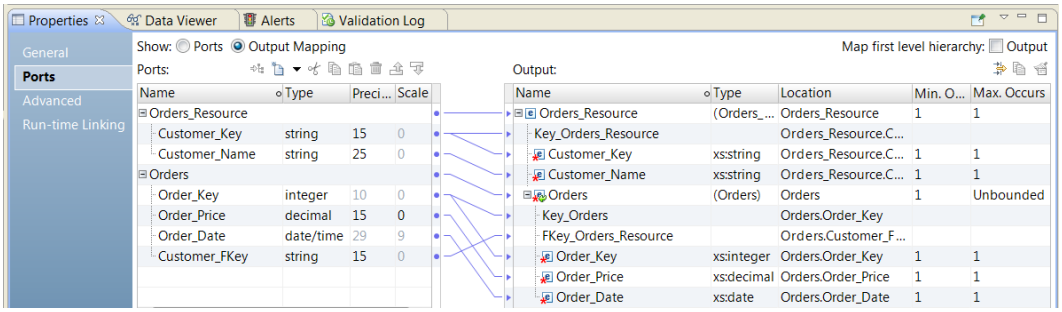
   Note: You cannot drag a port to create a link.

3. Select the Order_Key port and the Key_Orders element. Click **Map** to create the link.

   Order_Key must have links to the Key_Orders element and the Order_Key element.

4. Select the Customer_Fkey port and the Fkey_Orders_Resource element. Click **Map** to create the link.

   The following image shows the output mapping with the links to the keys:



Step 4. Test the Mapping in the Data Viewer View

You can run the REST resource mapping in the **Data Viewer** view. You can filter the output data by resource ID or by a filter condition in the **Data Viewer** view.

You can filter the data by entering values for elements from the Orders_Resource group. The elements are Customer_Key or Customer_Name. You cannot filter output data using elements from the Orders group. The Orders group is a child group.

You can filter the output data by the resource identifier, or you can enter a filter condition. The resource identifier is the key, which is the Customer_Key element. The filter condition is an expression that you enter in an expression editor. Use a filter condition to filter output data when you are not searching by key.

If you do not enter a resource identifier or a filter condition, the mapping returns all rows in the **Data Viewer** view.

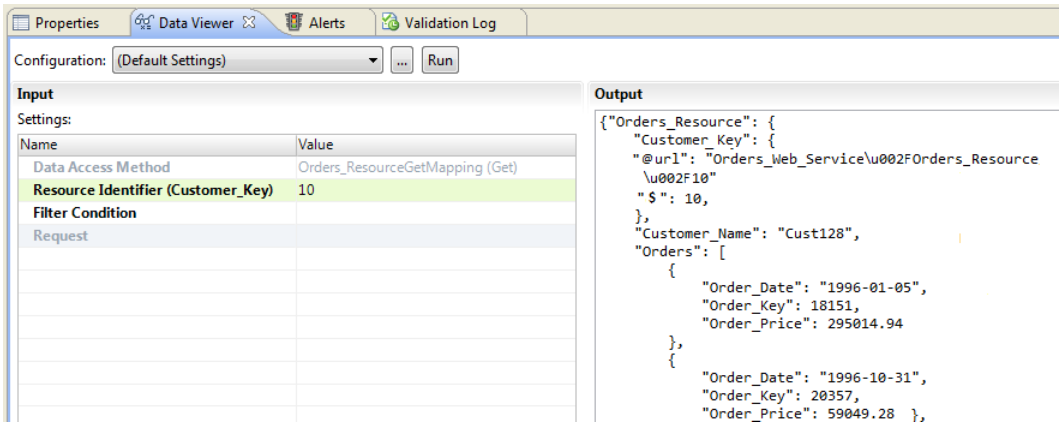
Filtering the Output by Resource ID

You can enter a resource ID to select output data by key values. The resource ID is the key that you selected when you created the resource. The resource ID is Customer_Key in this example.

1. In the **Data Viewer** view, enter a valid customer number in the Resource Identifier **Value** column.

2. Click **Run**.

   The output contains data for the customer that you entered.
   The following image shows the output for customer number 10:



Filtering the Output by Filter Condition

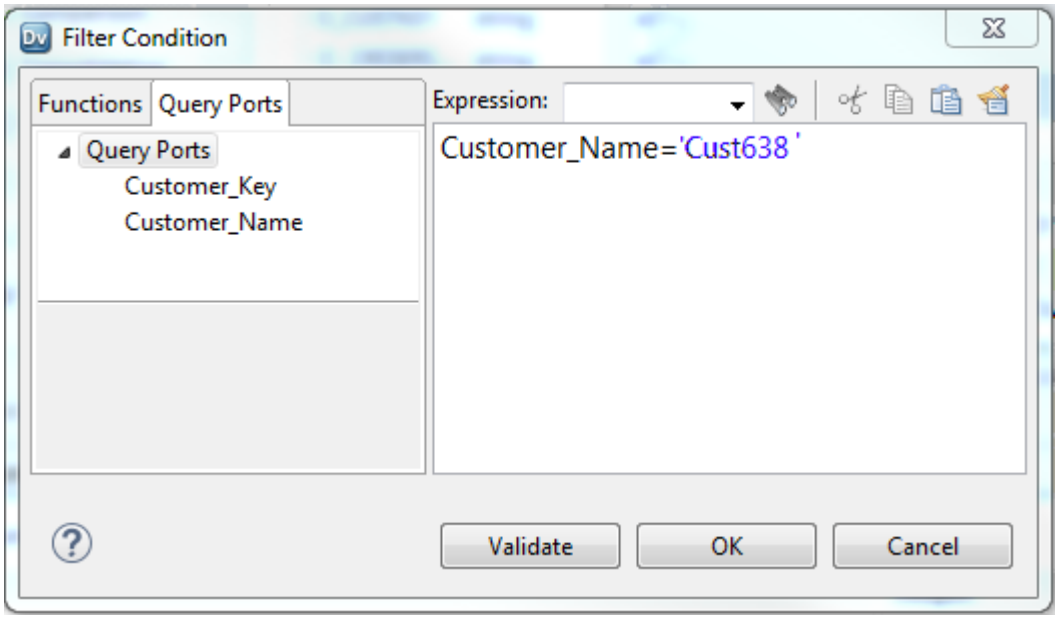
You can enter a filter condition to select output data by specific values.

1. In the **Data Viewer** view, click the selection arrow in the filter condition **Value** column.

   The Expression editor appears.

2. Click the **Query Ports** tab.

   A list of ports appears:



3. Select a query port and create a filter expression.

   You must enclose the search value in single quotes.

4. Click **Validate** to validate the expression.

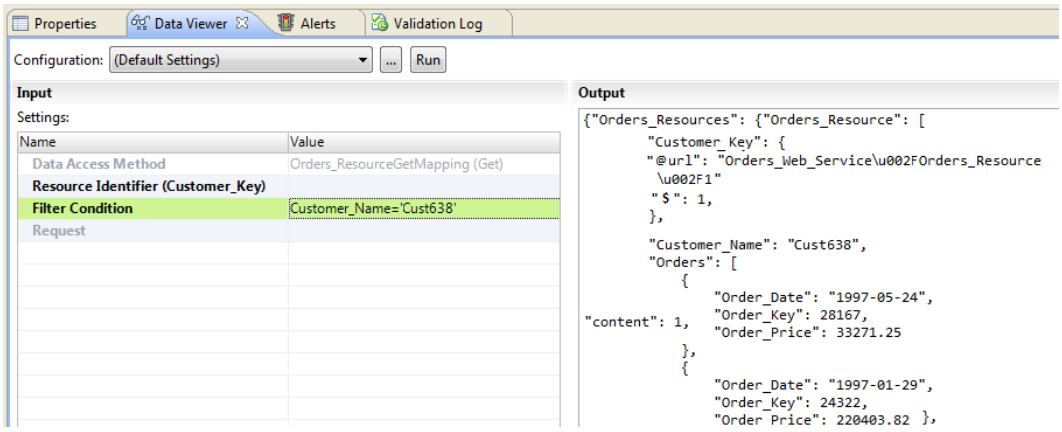
5. If the expression is valid, click **OK**.

   The expression appears as the filter condition in the **Data Viewer** view.

6. In the **Data Viewer** view, click **Run**.

   The orders for customer name Cust638 appear on the Output panel.

   The following image shows the results on the Output panel:



Step 5. Deploy the Application

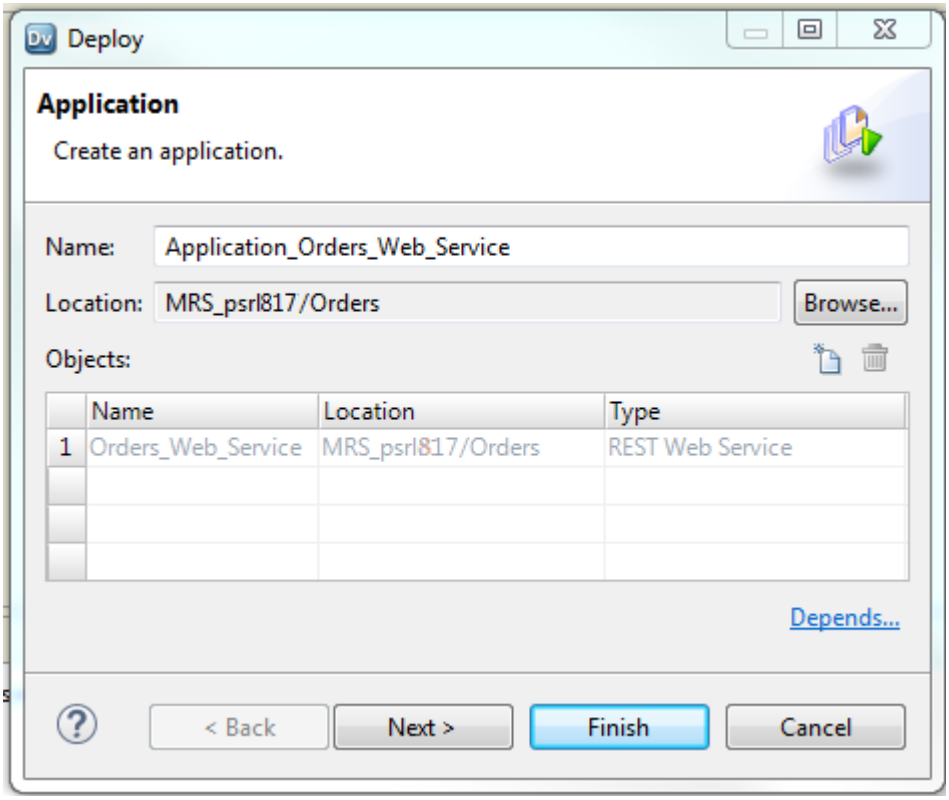
Deploy the web service as an application to a Data Integration Service. You must deploy the application to enable a web service client to connect to the web service.

After you deploy the web service, you can view the application in the Administrator tool.

Deploying the Application

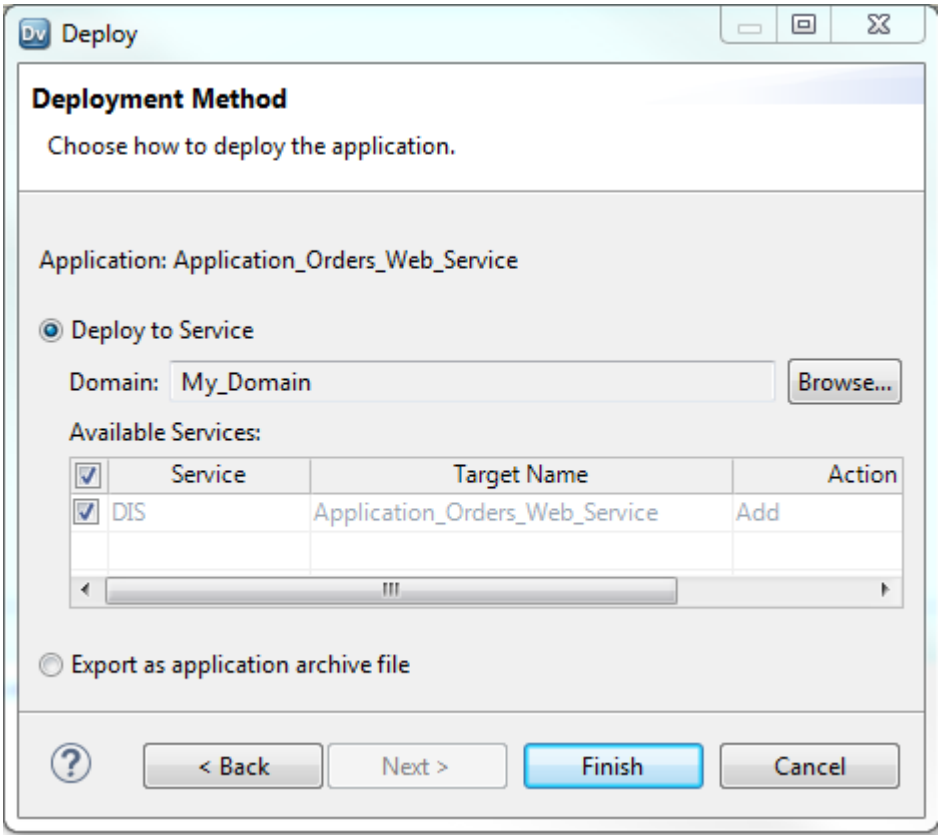
Deploy the application to a Data Integration Service.

1. In the **Object Explorer** view, right-click Orders_Web_Service and click **Deploy**:



2. Accept the default name and location. Click **Next**.

3. Choose the domain and the Data integration Service to deploy the application to.



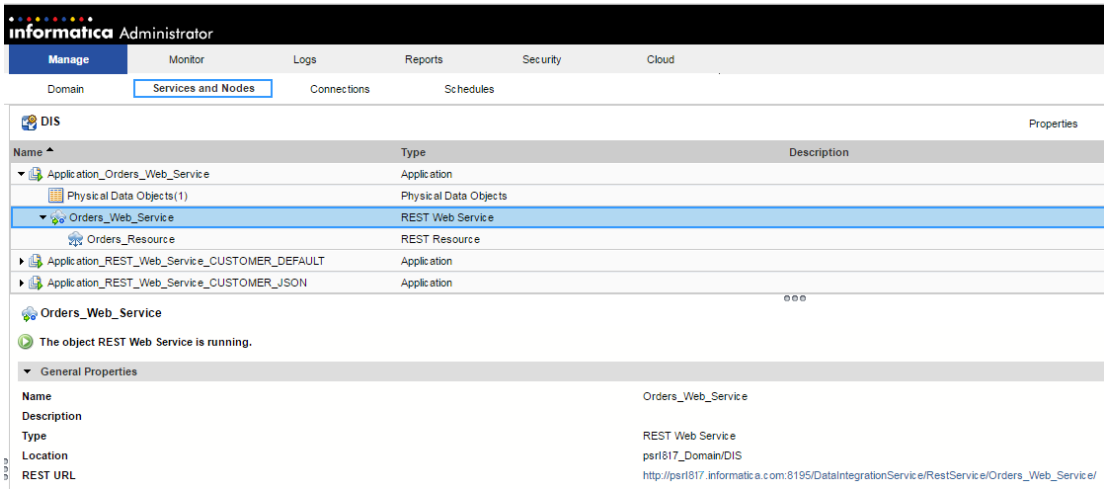
4. Click **Finish** to deploy the application.

Step 6. Query the Web Service from a Browser

Query the REST web service from a browser.

After you deploy the web service to a Data Integration Service, the web service appears as an application in the Administrator tool. You can view the URL to access the web service on the application **General Properties** panel.

The following image shows the REST URL on the **General Properties** panel for the web service:

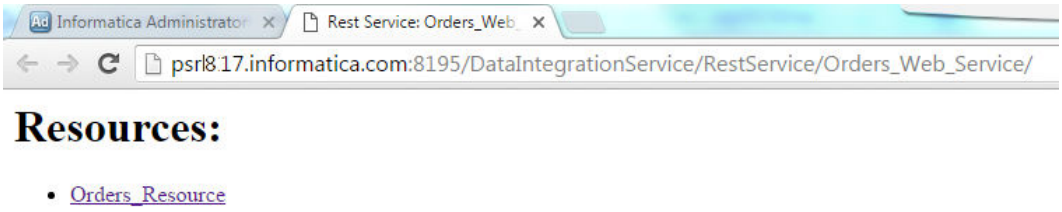


Querying the Web Service

You can query the web service by including a resource ID in the URL or by including a filter expression in the URL.

1. In the Administrator tool, click the REST URL on the web service application **General Properties** panel.

   The Data Integration Service lists the resources in the web service.
   The following image shows the resource in the Orders_Web_Service:

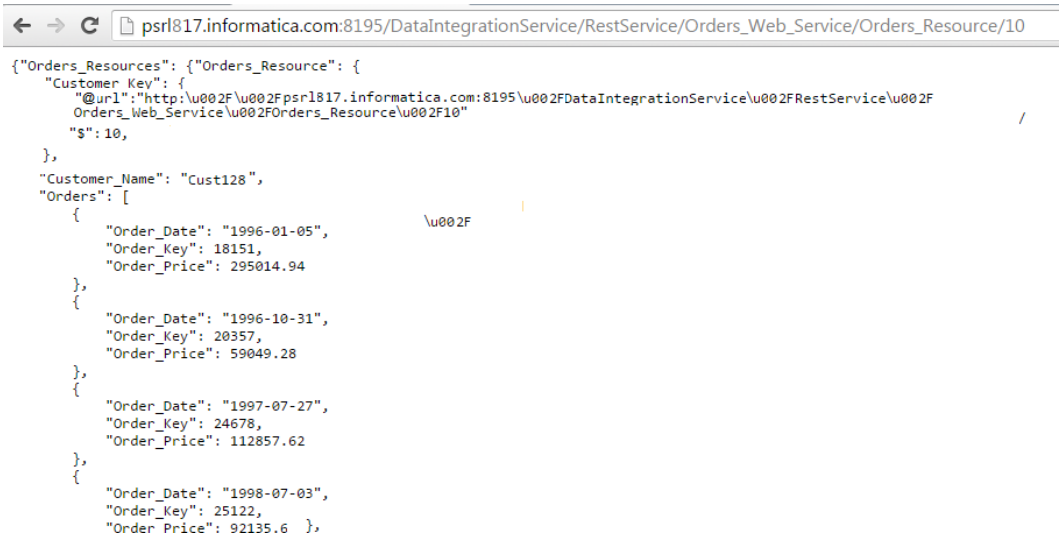


2. Click the **Orders_Resource** link.

   The web service returns all the output data from the resource mapping.

3. To search for a specific customer, add the resource name and the resource ID to the end of the URL in the browser. The link is `http://psrl817.informatica.com:8195/DataIntegrationService/RestService/Orders_Web_Service/Orders_Resource/10`.

   The following image shows the results of a query that searches for customer number 10 in the Orders_Resource:



4. To search by customer name instead of by the key, include a filter condition in the URL instead of the resource ID:

   `http://psrl817.informatica.com:8195/DataIntegrationService/RestService/Orders_Web_Service/Orders_Resource/?filterCondition=Customer_Name='Cust628'`

Authors

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